



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to ANNAMAR AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

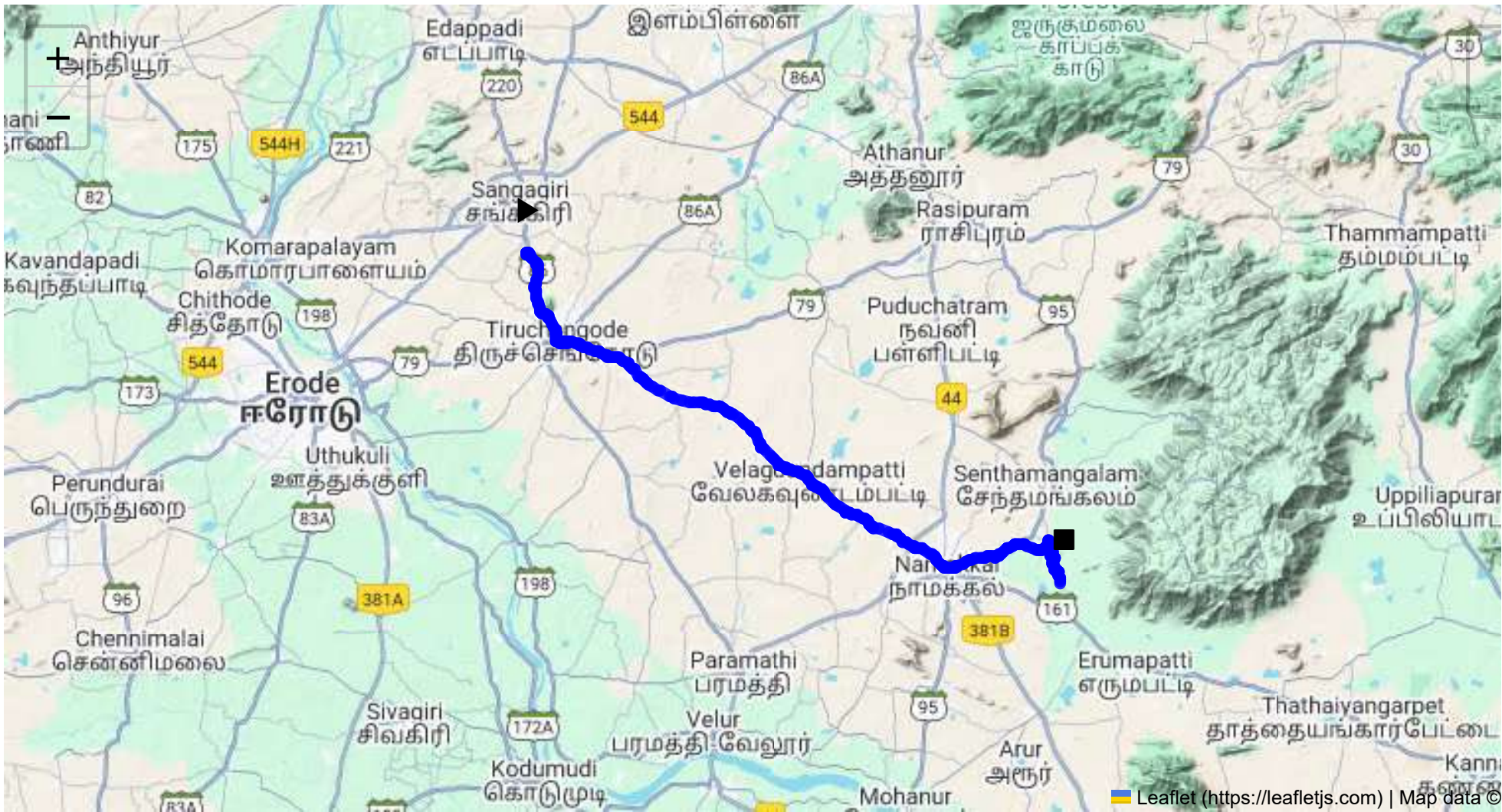
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 56.32 km
Estimated Duration: 1.3 hours
Adjusted Duration (Heavy Vehicle): 1.6 hours
Start: (11.4381, 77.8734)
End: (11.21373, 78.24691)



Welcome to the Journey Risk Management Study

1. Route Overview

The route from CVQF+23W, Sangagiri, Tamil Nadu, to 667W+FQ Sarkarpalayapalayam, via S Car St, Kamalar, Tiruchengode, spans approximately 56.32 kilometers. This path mostly follows major state roads and incorporates segments of rural and urban environments. Expect a mix of smooth highways and narrow town roads, with some winding sections typical of this region.

2. Weather Conditions and Hazards

The region experiences a tropical climate. Probable weather-related hazards include:

- **Monsoon Season (June-September):** Heavy rain can lead to flooding and reduced visibility. Roads might be slippery.
- **Summer (March-May):** High temperatures can lead to discomfort and road surface deterioration. Ensure good tire condition and avoid travel during severe weather warnings.

3. Traffic Patterns

- **Peak Hours:** Mornings (7-9 AM) and Evenings (5-7 PM) see increased local traffic, especially near towns such as Tiruchengode.
- **Congestion-Prone Areas:** Near market areas in Tiruchengode and at intersections along major highways.
- Plan travel avoiding peak hours for smoother journeys.

4. Road Quality and Infrastructure

- Major highways are generally in good condition with regular maintenance.
- Some rural roads might have potholes and require cautious navigation.
- Urban sections could have narrow lanes and abrupt turns.

5. Alternative Routes for Emergencies

- **Route 1:** Using SH-86 and bypassing Tiruchengode could be considered, though it may add distance.
- **Route 2:** Diverting through local bypass roads during urban congestion could be a practical alternative.

6. Local Regulations on Hazardous Material

- Ensure compliance with Tamil Nadu's transport regulations for hazardous materials, focusing on timings and documentation.
- Restrictions might include prohibitions on hazardous materials in certain urban areas and periods of high traffic.

7. Historical Incidents

There are occasional reports of truck accidents due to overloading or loss of control, typically in higher traffic areas or during heavy rains. No major disasters involving hazardous materials recorded recently along this specific route.

8. Environmental Considerations

- **Sensitive Areas:** Be cautious near agricultural zones, especially during planting or harvest seasons.

- Avoid any spillage and ensure trucks are not leaking hazardous materials to protect local environments.

9. Communication Coverage

Most of the route is covered by a good telecom network, but some rural stretches between Sangagiri and Tiruchengode may experience reduced signal strength.

10. Emergency Response Times

- **Urban Areas (Tiruchengode):** Emergency services can respond within 20-30 minutes.
- **Rural sections:** Response times may increase up to 45-60 minutes, depending on accessibility.

11. Overall Risk Assessment Summary

- **Moderate Risk:** Organization and adherence to traffic regulations can ensure safe transit.
- Key risks include adverse weather conditions, traffic congestion, and road quality in rural areas.
- Proper planning, timely vehicle checks, and constant communication are essential to mitigate these risks.

In conclusion, while the route presents common challenges related to rural and urban driving, these can be managed through caution, pre-trip inspections, and adjustments based on weather and traffic reports. Ensure compliance with local transport regulations and maintain continuous communication throughout the journey to ensure safety and efficiency.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Turn	High	11.37868, 77.89498	15 KM/Hr
5	Turn	Medium	11.37809, 77.89834	30 KM/Hr
6	Turn	Medium	11.23746, 78.15041	30 KM/Hr
7	Turn	Medium	11.22410, 78.17210	30 KM/Hr
8	Blind Spot	Blind Spot	11.24309, 78.23751	10 KM/Hr
9	Turn	High	11.23967, 78.23801	15 KM/Hr
10	Turn	Medium	11.23939, 78.23885	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
11	Turn	Medium	11.23898, 78.23928	30 KM/Hr
12	Turn	High	11.23506, 78.24036	15 KM/Hr
13	Turn	Medium	11.23303, 78.23971	30 KM/Hr
14	Turn	Medium	11.23265, 78.23980	30 KM/Hr
15	Turn	High	11.23086, 78.24171	15 KM/Hr
16	Turn	Medium	11.22239, 78.24264	30 KM/Hr
17	Turn	Medium	11.22141, 78.24463	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
11	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
13	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium
18	hospital	Aruthra Hospital	11.2264666, 78.1631524	30 km/h	Medium
19	clinic	Sri Ram Nursing Home, Namakkal	11.2259986, 78.1631944	30 km/h	Medium
20	hospital	Rudhra Hospital	11.226569, 78.163216	30 km/h	Medium
21	hospital	Annai Eye Hospital	11.2241645, 78.1641366	30 km/h	Medium
24	clinic	Chandra Test Tube Baby Centre	11.2241806, 78.1645931	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
25	clinic	Thangavel Skin Care	11.2253765, 78.1644957	30 km/h	Medium
26	hospital	Rajendran Children's Hospital	11.2251887, 78.1649142	30 km/h	Medium
27	clinic	Lung Care Centre	11.2276789, 78.1638586	30 km/h	Medium
28	hospital	Vignesh Hospital	11.224818, 78.1621315	30 km/h	Medium
29	clinic	Vishnu Ayurveda Clinic	11.224262, 78.164173	30 km/h	Medium
30	clinic	Dhanvanthiri Physiotheraphy	11.2245256, 78.1638528	30 km/h	Medium
31	hospital	Venkat Hospital	11.2253645, 78.1646159	30 km/h	Medium
32	hospital	Shree Meenakshi Dental Hospital	11.225269, 78.164275	30 km/h	Medium
34	clinic	Kumaran Poly Clinic	11.2233138, 78.1639212	30 km/h	Medium
35	hospital	Aravinth Hospital	11.2233336, 78.1672805	30 km/h	Medium
36	hospital	Ilamurugan Hospital	11.2248399, 78.166586	30 km/h	Medium
37	hospital	Kathiravan Hospital	11.226651, 78.167351	30 km/h	Medium
38	hospital	Aburvaa Ayurveda Hospital	11.2228807, 78.1640453	30 km/h	Medium
39	hospital	Vanitha Child Care & Skin Hospital	11.2243378, 78.16811	30 km/h	Medium
40	hospital	Sri Dhanvanthries Hospital	11.2241181, 78.1678614	30 km/h	Medium
41	hospital	Sri Nidhi Hospital, Namakkal	11.2255711, 78.1687291	30 km/h	Medium
42	clinic	Athithya Nursing Home	11.2237726, 78.1686447	30 km/h	Medium
43	hospital	Sri Ramana Hospital	11.223503, 78.1685433	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
9	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
10	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
12	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
14	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
15	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
16	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
17	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium
22	school	Government Elementary School	11.224555, 78.163967	30 km/h	Medium
23	school	Government Secondary School	11.2244426, 78.1647533	30 km/h	Medium
33	school	Government Girls School	11.2238954, 78.1662583	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.37809, 77.89834



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.23746, 78.15041



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.22410, 78.17210



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.24309, 78.23751



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.23967, 78.23801



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.23939, 78.23885



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.23898, 78.23928



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.23506, 78.24036



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.23265, 78.23980



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.23086, 78.24171



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.22239, 78.24264



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.22141, 78.24463

